

PW EARNERS
DSA in C++

Daily Practice Problems (DPP) — 8
Topic: Binary Search

Multiple Choice Questions (Predict the Output)

For Q1 to Q5: Select the exact output or required property of each program snippet. Assume all required headers like `<iostream>`, `<vector>`, and `<algorithm>` are included, and `using namespace std;` is present.

Q1. What is the worst-case Time Complexity of the following `solve` function in terms of N ?

```
1 void solve(int n) {
2     int i = 1;
3     while (i < n) {
4         int j = n;
5         while (j > 0) {
6             j = j / 2;
7         }
8         i = i * 2;
9     }
10 }
```

- A. $O(N)$ B. $O(N \log N)$ C. $O(\log^2 N)$ D. $O(\log N)$

Q2.

```
1 int main(){
2     vector<int> v = {1, 2, 4, 4, 4, 6, 7};
3     auto l = lower_bound(v.begin(), v.end(), 4);
4     auto u = upper_bound(v.begin(), v.end(), 4);
5     cout << u - l;
6 }
```

- A. 2 B. 3 C. 4 D. 0

Q3.

```
1 int main(){
2     int arr[] = {2, 4, 6, 8, 10};
3     int l = 0, r = 4;
4     while (l < r) {
5         int mid = l + (r - l) / 2;
6         if (arr[mid] < 5) l = mid;
7         else r = mid - 1;
8     }
9     cout << l;
10 }
```

- A. 1
B. 2
C. Infinite Loop (TLE)
D. Compilation Error

Q4.

```
1 int main(){
2     int arr[] = {10, 20, 30, 40, 50};
3     cout << *lower_bound(arr, arr + 5, 35);
4 }
```

- A. 30
B. 35
C. 40
D. Garbage Value

Q5. Crazy Question

```
1 int main(){
2     vector<int> v = {50, 40, 30, 20, 10};
3     auto it = lower_bound(v.begin(), v.end(), 30);
4
5     if(it != v.end()) cout << *it;
6     else cout << "-1";
7 }
```

- A. 30
B. -1
C. Undefined Behavior / Unpredictable
D. Compilation Error

For Q6 to Q15: Write clean, optimal, and compilable C++ programs. Assume inputs are provided by the user. Ensure your time complexities are optimal, leveraging Binary Search logic.

Q6. Find Smallest Letter Greater Than Target. Given a character array sorted in non-decreasing order and a character target, return the smallest character in the array that is strictly greater than the target. The letters wrap around (e.g., if target is 'z' and 'a' is in the array, return 'a'). Solve in $O(\log N)$.

Constraints: $2 \leq N \leq 10^4$.

Sample Input: letters = ['c', 'f', 'j'], target = 'a'

Sample Output: 'c'

Q7. Minimum in Rotated Sorted Array. Suppose an array of length N sorted in ascending order is rotated between 1 and N times. Given the rotated array of unique elements, return the minimum element of this array in $O(\log N)$ time.

Constraints: $1 \leq N \leq 5000$, $-5000 \leq arr[i] \leq 5000$.

Sample Input: {3, 4, 5, 1, 2}

Sample Output: 1

Q8. Rotation Count. Given an array of distinct integers sorted in increasing order and then rotated by an unknown amount, write a function to find out how many times the array was rotated. (Hint: The index of the minimum element relates to the number of rotations).

Constraints: $1 \leq N \leq 10^5$.

Sample Input: {15, 18, 2, 3, 6, 12}

Sample Output: 2 (The array is rotated right 2 times)

Q9. Frequency of Target Element. Given a sorted array of N integers and a target element, find the total number of occurrences of the target in the array. You must achieve $O(\log N)$ time complexity.

Constraints: $1 \leq N \leq 10^5$.

Sample Input: `arr = {1, 1, 2, 2, 2, 2, 3}, target = 2`

Sample Output: 4

Q10. Check Majority Element. Given a sorted array of N elements and a target element X , check if X is the majority element (i.e., it appears more than $N/2$ times in the array). Return `true` or `false`. Solve optimally in $O(\log N)$.

Constraints: $1 \leq N \leq 10^4$.

Sample Input: `arr = {1, 2, 3, 3, 3, 3, 10}, X = 3`

Sample Output: `true`

Q11. First 1 in an Infinite Binary Array. You are given an infinite size sorted binary array containing only 0s followed by 1s. Since the array is infinite, you cannot use `arr.size()`. Find the index of the first occurrence of 1.

Constraints: The first '1' will be within index 10^6 .

Sample Input: `arr = {0, 0, 0, 0, 1, 1, 1...}`

Sample Output: 4

Q12. Missing Number in Arithmetic Progression. Given an array that represents elements of an Arithmetic Progression in order, but one element is missing. Find the missing element in $O(\log N)$ time. Assume the first and last elements are never missing.

Constraints: $3 \leq N \leq 10^5$.

Sample Input: `arr = {2, 4, 8, 10, 12, 14}`

Sample Output: 6

Q13. Floor and Ceil in Sorted Array. Given a sorted array of N integers and an integer X , write a function that returns both the floor and the ceiling of X in the array. The floor of X is the greatest element smaller than or equal to X , and the ceiling is the smallest element greater than or equal to X . Return -1 for either if they don't exist.

Constraints: $1 \leq N \leq 10^5$.

Sample Input: `arr = {1, 2, 8, 10, 10, 12, 19}, X = 5`

Sample Output: Floor = 2, Ceil = 8

Q14. Search in a Bitonic Array. A Bitonic array is an array that is strictly increasing and then strictly decreasing. Given a bitonic array of distinct integers and a target integer, return the index of the target in $O(\log N)$ time. Return -1 if the target is not found.

Constraints: $3 \leq N \leq 10^4$.

Sample Input: `arr = {1, 3, 8, 12, 4, 2}, target = 4`

Sample Output: 4 (index of element 4)

Q15. K-th Element of Two Sorted Arrays. Given two sorted arrays of sizes M and N respectively, you are tasked with finding the element that would be at the k^{th} position of the final sorted array. Solve this optimally with a Binary Search approach in $O(\log(\min(M, N)))$ time complexity.

Constraints: $1 \leq M, N \leq 10^4$, $1 \leq k \leq M + N$.

Sample Input: `arr1 = {2, 3, 6, 7, 9}, arr2 = {1, 4, 8, 10}, k = 5`

Sample Output: 6 (Merged array is {1, 2, 3, 4, 6, 7, 8, 9, 10})

All the best! Keep Coding ♡